

TEMPEST: Porewater SO₄/Cl

202606 142-221 Samples

2026-09-11

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Sample_Year = "2026"
Date_Run = "2026-09-11" #Date that instrument was run
Run_by = "Zoe Read" #Instrument user
Script_run_by = "Zoe Read" #Code user
run_notes = "TMP_FW_R0_S04_20260610_D4_T2 has high Cl for FW so might rerun it." #any notes from the

##Sample data that was entered incorrectly
# The Old ID is the original, incorrectly-entered ID and the New ID is the correct ID to change it to.
Old_ID_1 = NA
New_ID_1 = NA

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/COMPASS_TEMPEST_202606_142-221_Cl.txt"
raw_file_name_so4 = "Raw Data/COMPASS_TEMPEST_202606_142-221_S04.txt"

#Sample ID Path
SampleID_path = "Raw Data/TEMPEST_S04_20260501-20260612_corrected.csv"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_TEMPEST_Processed_Cl_S04_202606_142-221.csv"

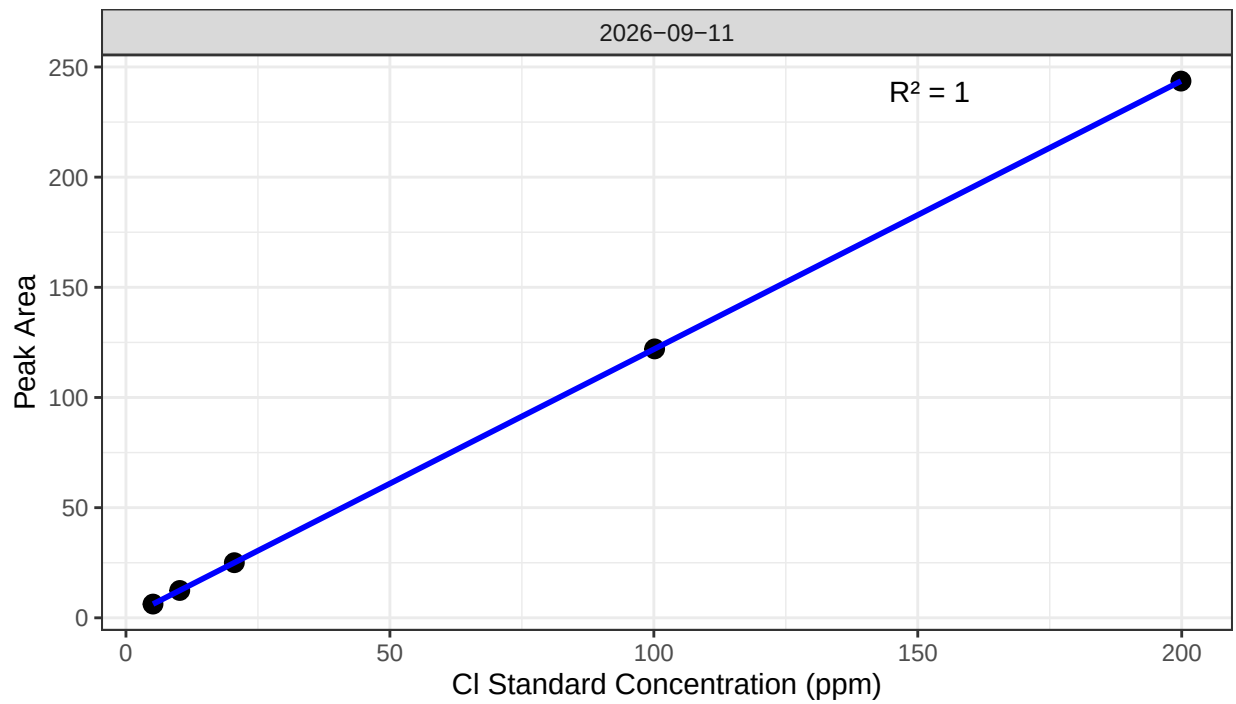
##### Log Files - PLEASE CHECK

#qaqc log file path for this year copied over from COMPASS GitHub
Log_path = "Raw Data/COMPASS_Synoptic_Cl_S04_QAQClog_2026.csv"

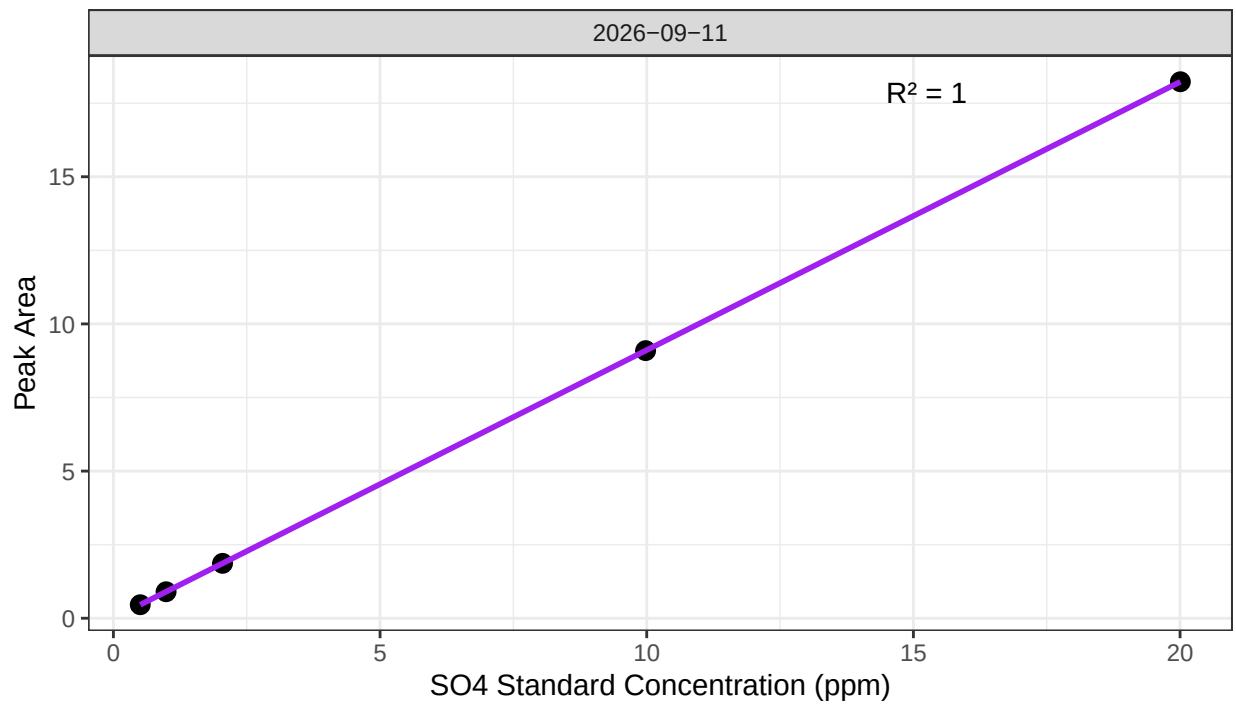
##Set Up Code - constants and QAQC cutoffs
##Import Sample Data
##Fix Sample IDs entered wrong
```

0.2 Assess Standard Curves

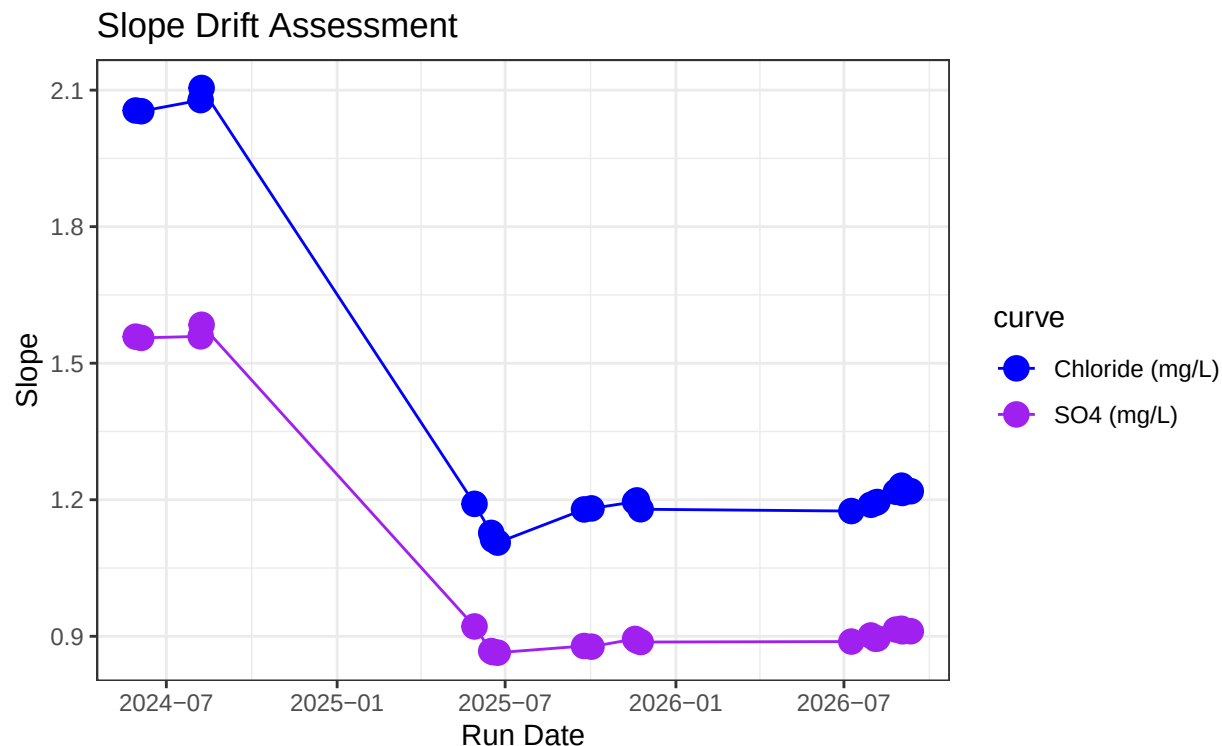
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

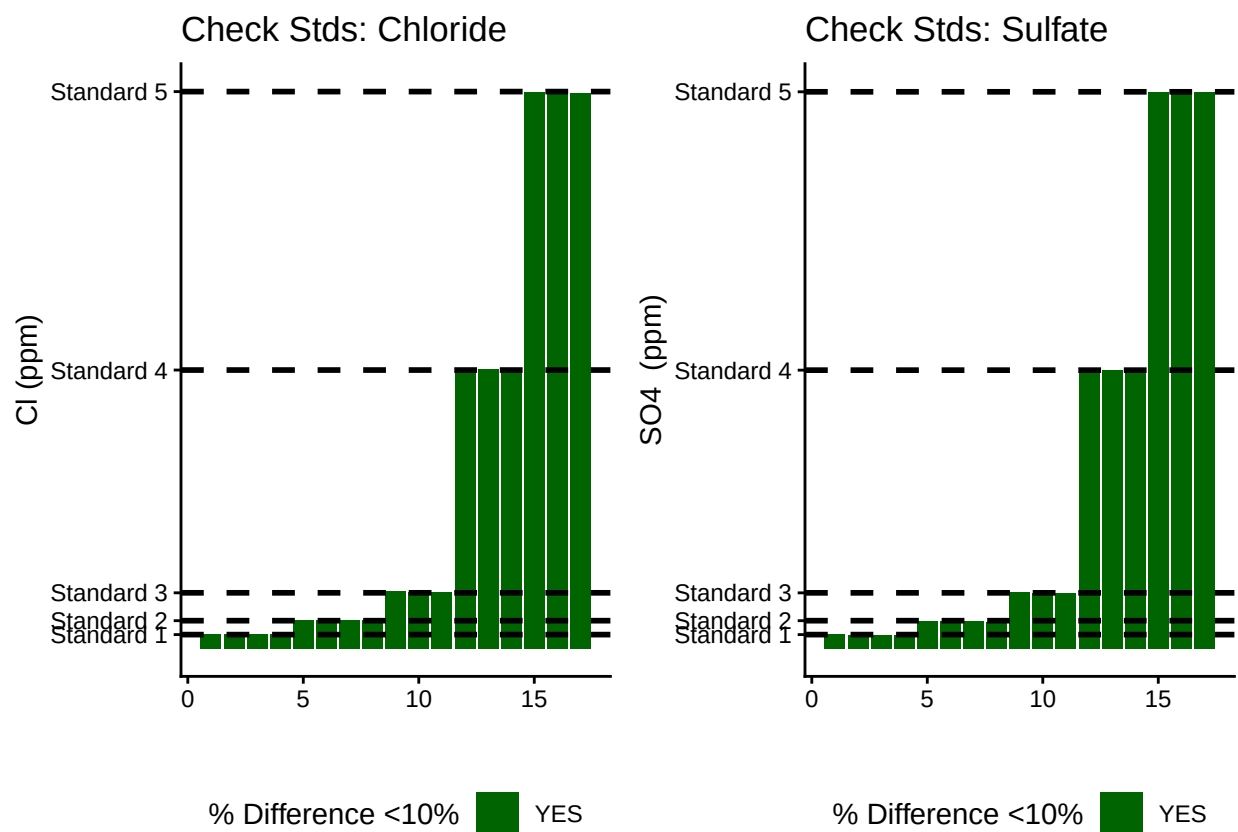
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  5.09 0.0341 0.00669 Chloride Check Standard RSD within Range --
## 2 Standard 2 10.1 0.0599 0.00591 Chloride Check Standard RSD within Range --
## 3 Standard 3 20.4 0.127 0.00624 Chloride Check Standard RSD within Range --
## 4 Standard 4 100. 0.0501 0.000500 Chloride Check Standard RSD within Range --
## 5 Standard 5 200. 0.229 0.00115 Chloride Check Standard RSD within Range --
```

```
## # A tibble: 5 x 5
##   sample_ID mean_S04 sd_S04 cv_S04 flag_S04
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  0.489 0.0103 0.0210 Sulfate Check Standard RSD within Rang~
## 2 Standard 2  0.977 0.0119 0.0122 Sulfate Check Standard RSD within Rang~
## 3 Standard 3  2.00 0.0379 0.0189 Sulfate Check Standard RSD within Rang~
## 4 Standard 4  9.99 0.0139 0.00139 Sulfate Check Standard RSD within Rang~
## 5 Standard 5 20.0 0.000800 0.0000400 Sulfate Check Standard RSD within Rang~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



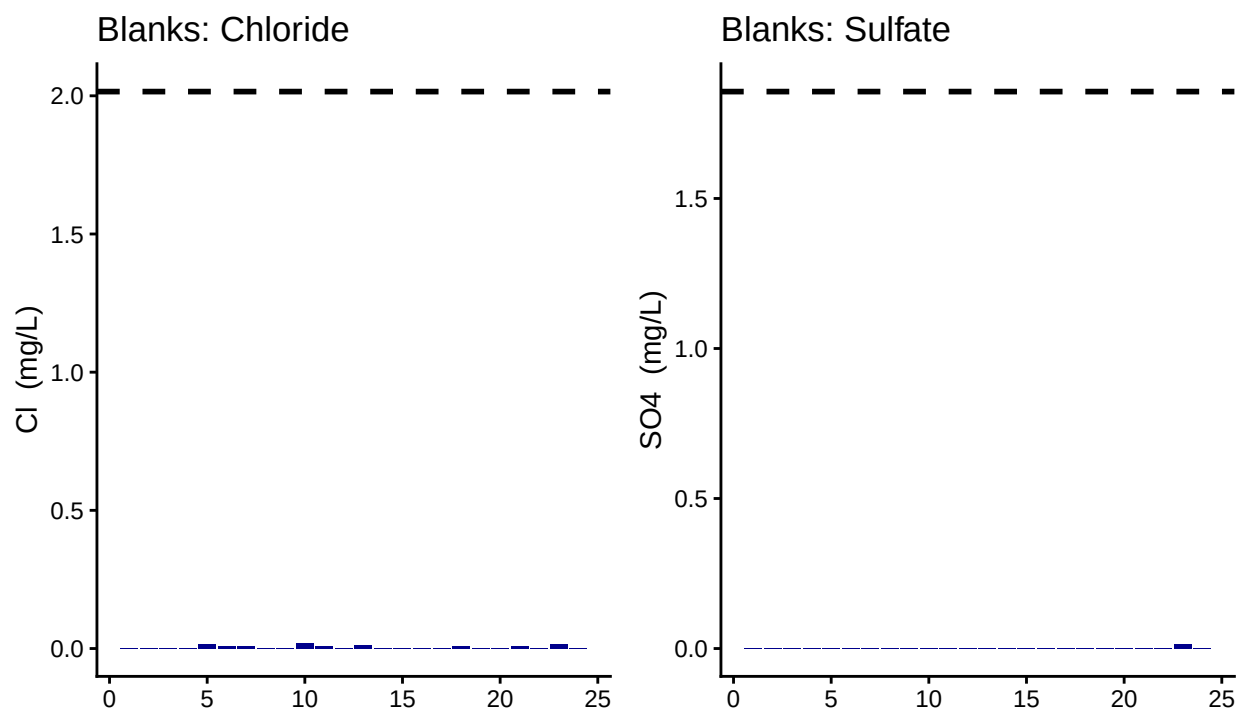
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples YE Blank Conc <25% Quartile Samples YE

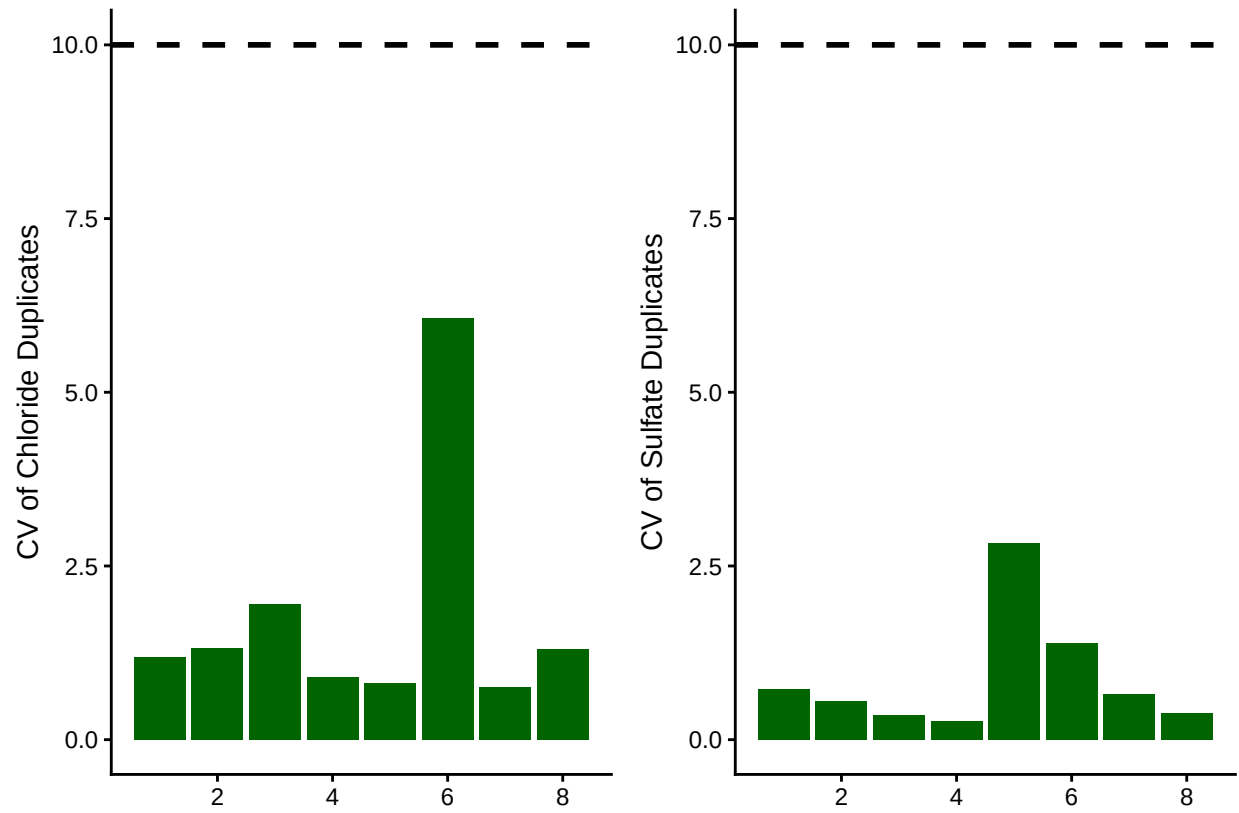
```
## Chloride blanks mean ppm:
```

```
## [1] 0.004458333
```

```
## Sulfate blanks mean ppm:
```

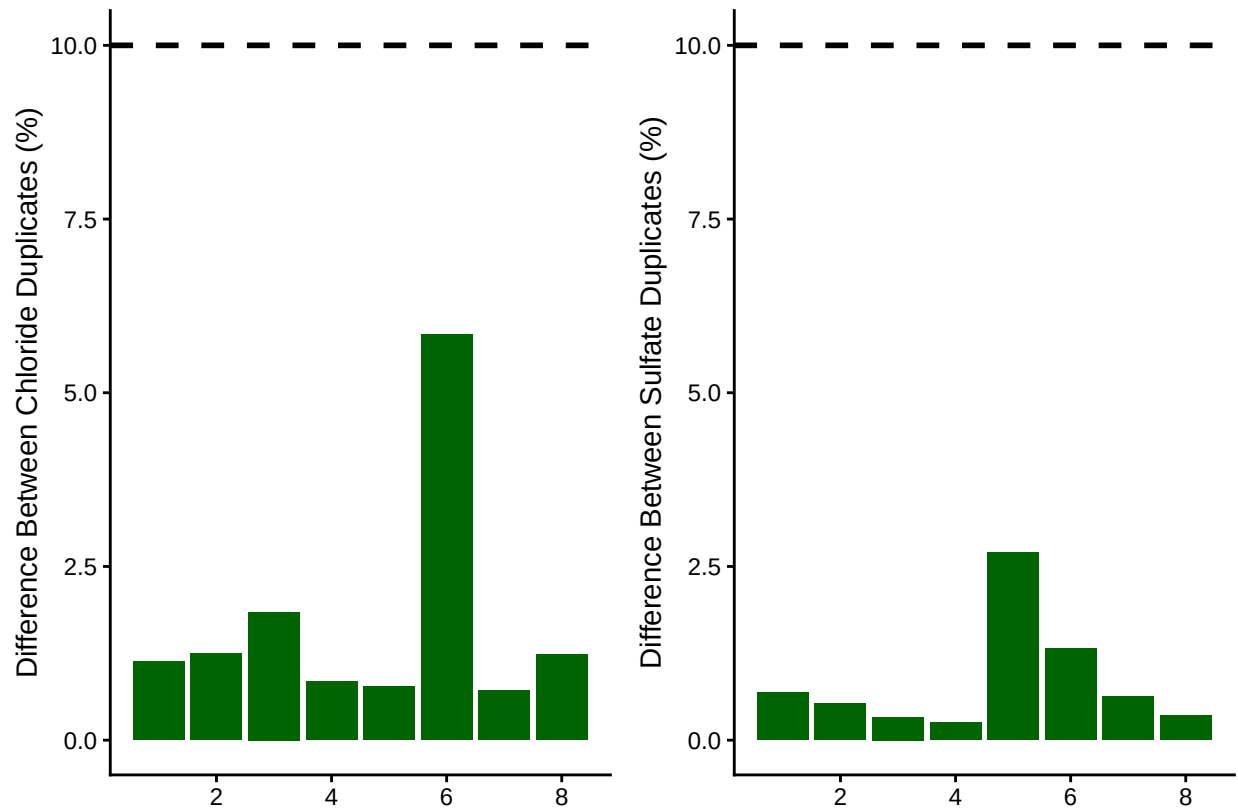
```
## [1] 0.0006541667
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for TEMPEST this depends on the sample so...
all_dat$Dilution <- 50
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "TMP"), 50, all_dat$Dilution)
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "EST_SourceWater"), 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "SW_SourceWater"), 100, all_dat$Dilution)
```

```

all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "FATE"), 100, all_dat$Dilution)

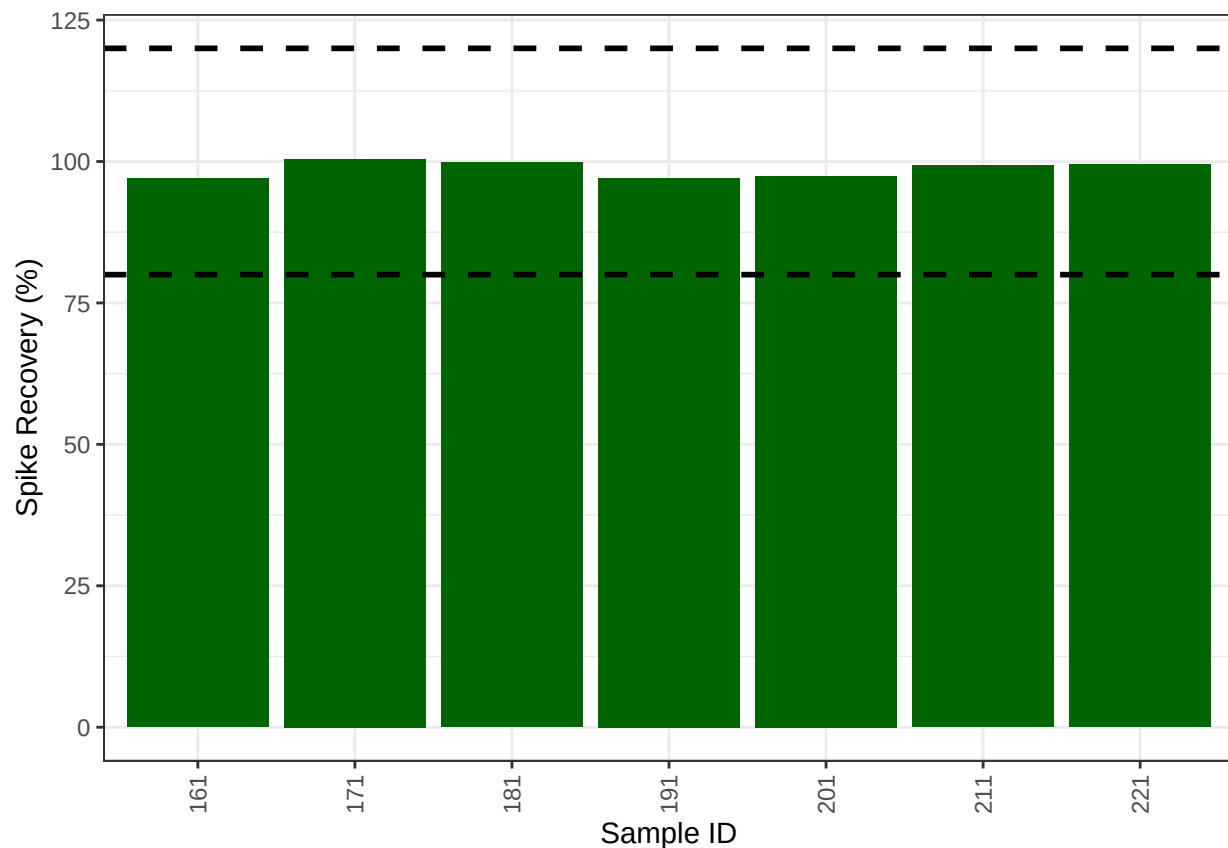
all_dat$Dilution <- ifelse(all_dat$sample_ID == "90", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "92", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "93", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "94", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "129", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "132", 100, all_dat$Dilution)

all_dat$Dilution <- ifelse(all_dat$sample_ID == "169", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "171", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "172", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "205", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "207", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "208", 100, all_dat$Dilution)

# head(all_dat)

```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

Sample Flagging

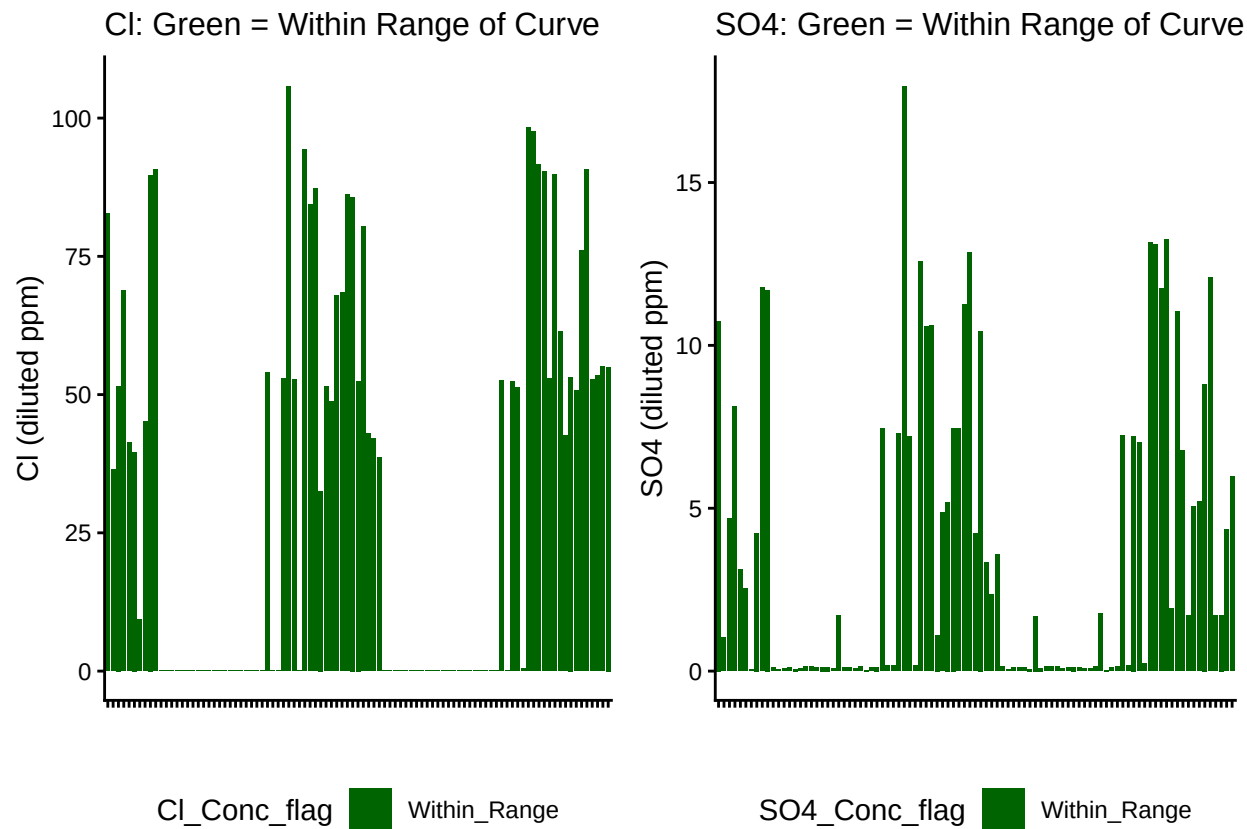


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

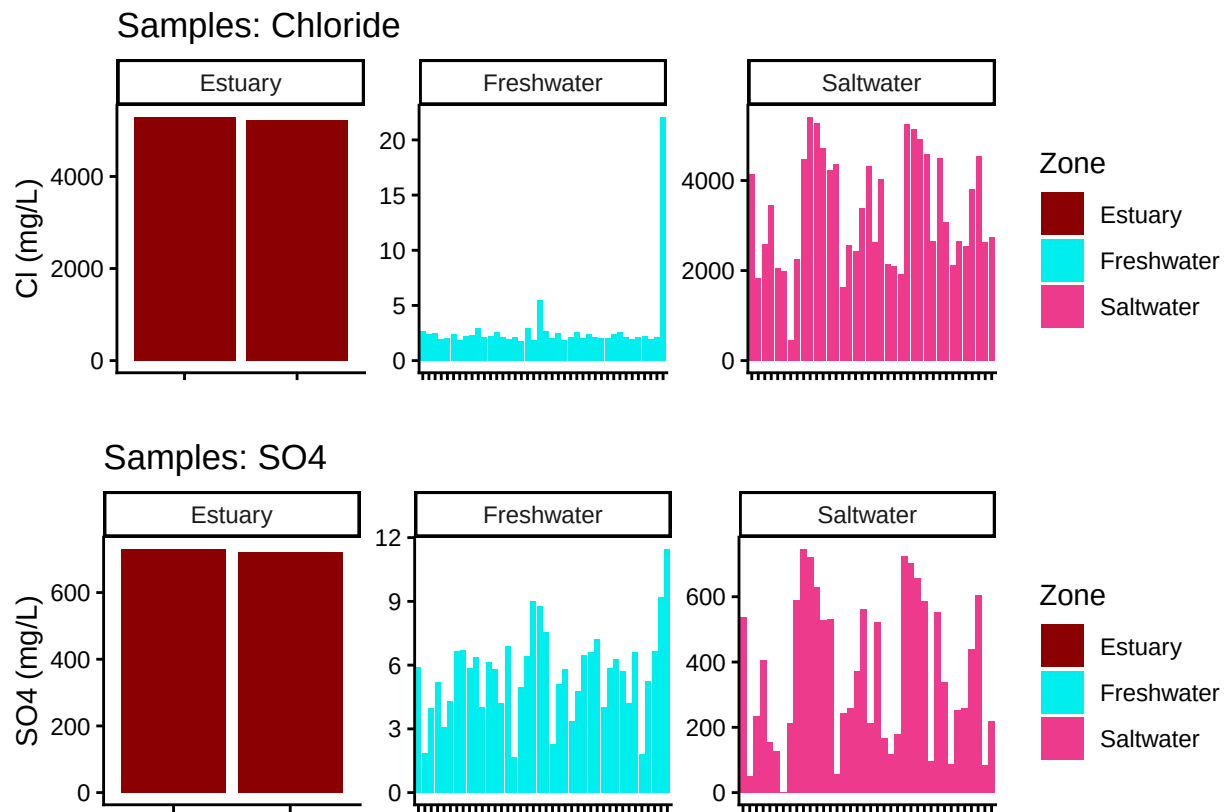
##Merge with Sample ID list to get actual sample IDs

##Pull in active porewater inventory

##Check samples against metadata

All sample IDs are present in metadata.

0.9 Visualize Data by Plot



0.10 Export Processed Data

#end